

ENARXI INNOVATIONS PRIVATE LIMITED

INTERNSHIP - PROJECT

WATER REFILLING SYSTEM

REPORT

1. Introduction

The **Water Refilling System** automates water tank refilling using an ESP32 microcontroller, sensors, a motor, and a user interface. It supports three operation modes: **Scheduled**, **Automatic (Auto)**, and **Manual**, with configuration via buttons and Bluetooth. This manual guides you through setup, operation, configuration, and troubleshooting.

System Components

- **ESP32 Microcontroller:** Controls the system.
- **Sensors:**
 - Dry Run Sensor (GPIO 5): Detects water presence (LOW = water present).
 - Float Sensor (GPIO 4): Detects tank full (LOW = full).
- **Motor:** Controlled via relay (GPIO 23, assumed active-high).
- **Buttons:**
 - Schedule (GPIO 16): Toggles Scheduled/Manual motor(2 Way Application).
 - Auto/Manual (GPIO 17): Switches between Auto and Manual modes.
 - Reset (GPIO 18): Resets system to default.
- **LEDs:**
 - Single-color: Water (GPIO 25), Tank Full (GPIO 26), Motor (GPIO 27), Manual (GPIO 19), Scheduled (GPIO 33).
- **LCD (I2C, 0x27):** Displays time, mode, status, and events.
- **RTC (I2C,0x57,0x68):** Provides time for Scheduled Mode.
- **Bluetooth:** Connects as "Water Refilling System" for configuration.
- **Serial:** USB interface at 115200 baud.

2. Setup

Hardware Requirements

- ESP32 board (Dev).
- DS3231 RTC module.
- 16x2 LCD with I2C.
- Dry run and float sensors .
- Relay module for motor control (5V, compatible with GPIO 23).

- Three pushbuttons (GPIO 16–18).
- Five single-color LEDs (GPIO 25–27, 19, 33).
- Power supply (5V for ESP32, suitable voltage for motor).

Installation

1. **Connect Hardware:**
 - Wire sensors, relay, buttons, LEDs, RTC, and LCD as per GPIO assignments.
 - Ensure proper voltage levels (e.g., 3.3V for ESP32 GPIO, 5V for relay).
2. **Power On:**
 - Connect ESP32 to a 5V power source via USB or external supply.
 - Ensure motor power is connected via relay.
3. **Install Software:**
 - Use Arduino IDE with ESP32 board support.
 - Install libraries: `RTClib`, `BluetoothSerial`, `LiquidCrystal_I2C`.
4. **Initial Display:**
 - LCD shows “ENARXI” and “WATER REFILL SYS” for 5 seconds, then current time (e.g., `12:00`) and `AUTO MOTOR OFF`.
 - Green LED (GPIO 32) lights up (Auto Mode default).

3. Operation Modes

The system operates in three modes, controlled via buttons or menu commands.

3.1 Scheduled Mode

- **Description:** Runs the motor at a set time (e.g., 06:00) every 1–2 days, ignoring sensors during the run. Post-run, the motor stays ON if water is present and the tank is not full; otherwise, it retries up to 3 times(default).
- **LCD Display:**
 - Row 1: `A HH:MM` (e.g., `A 06:00`).
 - Row 2: Default `XS-Y-ZM-WD` (e.g., `60S-3-1M-2D` = 60s run, 3 retries, 1min retry interval, 2-day interval).
 - Alternates with events (5s each): `Sch Run STARTED`, `Water Detected`, `Retry 1`, `Retry Wait Started`.
- **LEDs:** Scheduled (GPIO 33) ON, Water (25) and Tank Full (26) reflect sensor states, Motor (27) reflects motor state.
- **Activation:**
 - **Button:** In Auto Mode, hold the Schedule button (GPIO 16) for 3 seconds.(1st Case of using Schedule button)
 - **Menu:** Use Admin option 23 via Serial/Bluetooth.
 - **Note:** Requires RTC; if unavailable, displays `Sch Mode Disabled` and reverts to Auto.
- **Deactivation:** Repeat activation steps to toggle OFF (returns to Auto Mode).
- **Behavior:**
 - At alarm time, the motor runs for a set duration (1–60s or 1–5 min).

- Post-run: Motor ON if water present (GPIO 5 LOW) and tank not full (GPIO 4 LOW); else OFF ,number retries can be set from 0 to 5 with wait interval 1–60 minutes.
- The schedule can be set for everyday or alternate days by the user.
- Retries stop if water detected or retry count exceeded.

3.2 Automatic (Auto) Mode

- **Description:** Motor runs only when water is present and tank is not full, based on sensor inputs.
- **LCD Display:**
 - Row 1: Current time (e.g., 12:00).
 - Row 2: AUTO MOTOR ON (if water present and tank not full) or AUTO MOTOR OFF.
- **LEDs:** Scheduled and Manual are off so it indicates auto mode, Water (25) and Tank Full (26) reflect sensors, Motor (27) reflects motor.
- **Activation:**
 - System Default (Startup), It is in Auto Mode.
 - **Button:** Hold Auto/Manual button (GPIO 17) for 3 seconds (If in Manual Mode).
 - **Menu:** Automatically set after reset (GPIO 18 or Admin option 24).
- **Deactivation:** Switch to Manual Mode via Auto/Manual button.
- **Behavior:**
 - Motor ON if GPIO 5 LOW (water) and GPIO 4 LOW (not full).
 - Motor OFF otherwise.

3.3 Manual Mode

- **Description:** Motor state is manually toggled, ignoring sensors.
- **LCD Display:**
 - Row 1: Current time (e.g., 12:00).
 - Row 2: MANUAL MOTOR ON or MANUAL MOTOR OFF.
- **LEDs:** Manual (GPIO 19) ON, Water (25) and Tank Full (26) reflect sensors, Motor (27) reflects motor.
- **Activation:**
 - **Button:** Hold Auto/Manual button (GPIO 17) for 3 seconds to enter Manual Mode.(If in Auto Mode)
 - **Menu:** Use Admin option 22 to toggle motor state.
- **Motor Control:**
 - Hold the Schedule button (GPIO 16) for 3 seconds to toggle the motor ON/OFF. (2nd Case of using Schedule button)
- **Deactivation:** Switch to Auto Mode via Auto/Manual button.

4. Configuration

Settings are configured via Serial Monitor (115200 baud) or Bluetooth (“Water Refilling System”) using a terminal app (Serial Bluetooth Terminal). The menu is accessed by typing **M**.

4.1 Accessing the Menu

1. **Serial:**
 - Connect ESP32 to PC via USB.
 - Open Arduino IDE Serial Monitor (115200 baud, Newline or Carriage Return).
 - Type **M** and press Enter.
2. **Bluetooth:**
 - Pair with “Water Refilling System” on a phone/PC.
 - Use a terminal app, connect, and type **M** followed by Enter.

Menu Display:

=== Settings Menu ===

1. Alarm Time [06:00]
2. Run Duration [60 sec]
3. Retries [3]
4. Retry Interval [1 min]
5. Interval [2 days]
6. Status
7. Admin Menu
- M. Menu

4.2 Menu Options

- **1. Alarm Time:**
 - Enter **1**, then **HH:MM** (e.g., **06:00**).
 - Sets Scheduled Mode run time (00:00–23:59).
- **2. Run Duration:**
 - Scheduled On Motor run time ,in spite of considering sensor data
 - Enter **2**, then:
 - **11** for seconds (1–60), e.g., **10** for 10s.
 - **12** for minutes (1–5), e.g., **2** for 2min.
- **3. Retries:**
 - Enter **3**, then 0–5 (e.g., **2** for 2 retries).
 - Number of retry attempts in Scheduled Mode.
- **4. Retry Interval:**
 - Enter **4**, then 1–60 (e.g., **5** for 5 min).
 - Time between retries in Scheduled Mode.
- **5. Interval:**
 - Enter **5**, then 1–2 (e.g., **1** for daily).
 - Days between Scheduled runs.
- **6. Status:**

Enter **6** to view current state, e.g.:

Time: 12:00 | Water: YES | Tank: NO | Motor: OFF | Mode: AUTOMATIC

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- **7. Admin Menu:**
 - Enter **7**, then password **Enarxi0000**.
 - Options:
 - **21. Set RTC:** Enter **DD/MM/YYYY, HH:MM:SS** (e.g., **08/06/2025, 12:00:00**).
 - **22. Manual Mode:** Toggles motor ON/OFF in Manual Mode.
 - **23. Scheduled Mode:** Toggles Scheduled Mode ON/OFF.
 - **24. Reset:** Resets to default (Auto Mode, motor OFF).
- **M. Menu:** Returns to the main menu.

4.3 Default Settings

- Alarm Time: 06:00
- Run Duration: 60 seconds
- Retries: 3
- Retry Interval: 1 minute
- Interval: 2 days
- Mode: Auto

5. Button Controls

- **Auto/Manual Button (GPIO 17):**
 - Hold 3s to switch between Auto and Manual Modes.
- **Schedule Button (GPIO 16):**
 - In Auto Mode: Hold 3s to toggle Scheduled Mode.
 - In Manual Mode: Hold 3s to toggle motor ON/OFF.
- **Reset Button (GPIO 18):**
 - Hold 3s to reset to Auto Mode, motor OFF, default settings.
- **Note:** LCD shows **Process Initiated** during button press, then mode/status for 5s.

6. LED Indicators

- **Single-Color LEDs:**
 - Water (GPIO 25): ON if water is present (GPIO 5 LOW).
 - Tank Full (GPIO 26): ON if tank full (GPIO 4 HIGH).
 - Motor (GPIO 27): ON if motor is active.
 - Manual Mode (GPIO 19):
 - Scheduled Mode (GPIO 33):

7. LCD Displays

- **Startup:** ENARXI and WATER REFILL SYS for 5s.

	Row	Column Range	Context
	Row 0	5 - 10	Company Name
	Row 1	0 - 15	System Name

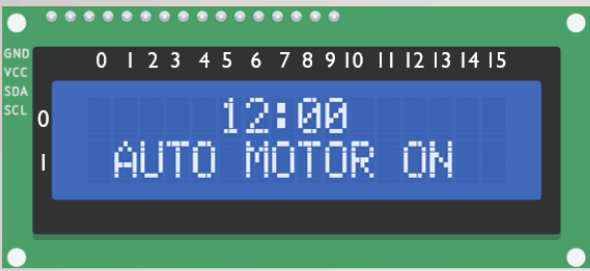
- **Scheduled Mode:**
 - Row 1: A HH:MM (e.g., A 06:00).
 - Row 2: XS-Y-ZM-WD (e.g., 60S-3-1M-2D) or events (Sch Run STARTED, Retry 1) every 5s.

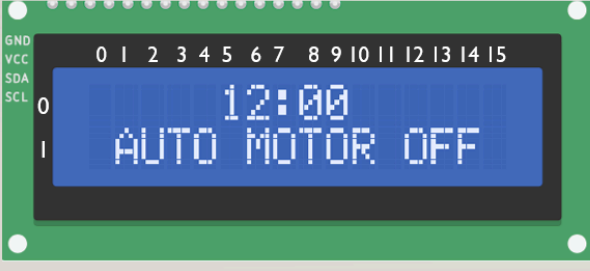
	Row	Column Range	Context
	Row 0	0 - 4 6 - 8 10 - 15	Current Time Operating mode Alarm Time with A
	Row 1	2 - 4 6 8 - 9 11 - 12	Motor Run Duration No of Retries Retry Wait Interval Interval Days

	Row	Column Range	Context
	Row 0	0 - 4 6 - 8 10 - 15	Current Time Operating mode Alarm Time with A
	Row 1	0 - 14	Operation Intimation

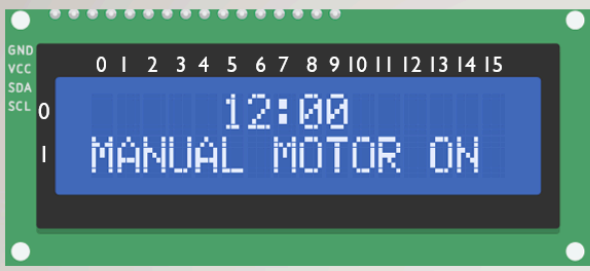
	Row	Column Range	Context
	Row 0	0 - 4 6 - 8 10 - 15	Current Time Operating mode Alarm Time with A
	Row 1	4 - 10	Operation Intimation

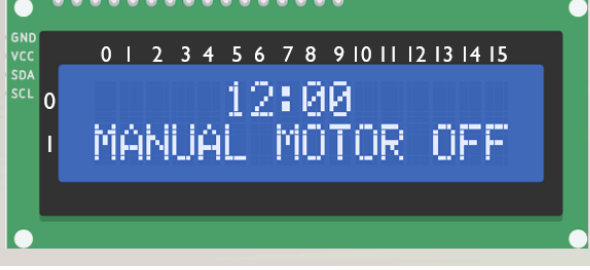
- **Auto Mode:**
 - Row 1: HH:MM (e.g., 12:00).
 - Row 2: AUTO MOTOR ON/OFF.

	Row	Column Range	Context
	Row 0	5 - 9	Current Time
	Row 1	1 - 4 6 - 13	Operating Mode Motor State

	Row	Column Range	Context
	Row 0	5 - 9	Current Time
	Row 1	1 - 4 6 - 14	Operating Mode Motor State

- **Manual Mode:**
 - Row 1: HH:MM.
 - Row 2: MANUAL MOTOR ON/OFF.

	Row	Column Range	Context
	Row 0	5 - 9	Current Time
	Row 1	0 - 5 7 - 14	Operating Mode Motor State

	Row	Column Range	Context
	Row 0	5 - 9	Current Time
	Row 1	0 - 5 7 - 15	Operating Mode Motor State

- **Errors/Events:** Temporary messages (5s), e.g., RTC Not Found, Invalid Option.

8. Troubleshooting

Issue	Possible Cause	Solution
LCD blank	Loose I2C wiring, wrong address	Check GPIO 21–22, verify address 0x27.

No Bluetooth	Not paired, wrong app	Pair with “Water Refilling System”, use the terminal app.
Motor not running	Relay issue, wrong mode	Check relay wiring (GPIO 23), ensure correct mode (Auto: water present, Manual: toggled ON).
RTC error (RTC Not Found)	RTC disconnected, low battery	Check I2C wiring, replace RTC battery.
Buttons unresponsive	Wiring issue, short hold	Hold 3s, check GPIO 16–18 wiring.
Wrong time	RTC not set	Use Admin option 21 to set time.
Menu not responding	Wrong baud, no newline	Use 115200 baud, enable Newline in Serial Monitor.

9. Safety Precautions

- **Power Supply:** Ensure correct voltage (3.3V for ESP32, appropriate for motor).
- **Water Exposure:** Protect electronics from water damage.
- **Relay Handling:** Verify relay is rated for motor current.
- **Testing:** Test motor control in Manual Mode before enabling Auto/Scheduled.

10. Specifications

- **Power:** 3.3V DC (ESP32), motor voltage as per relay.
- **Communication:** Serial (115200 baud), Bluetooth.
- **Sensors:** Dry run (active-low), Float (active-high).
- **Motor Control:** Relay (GPIO 23, assumed active-high).
- **Display:** 16x2 LCD (I2C, 0x27).
- **Timekeeping:** DS3231 RTC (I2C).

11. Contact Support

For issues not covered, contact the system provider with:

- Serial/Bluetooth logs.
- LCD photos/videos.
- Hardware setup details.
- Observed behavior vs. expected.

COMMAND LIST

Serial and Bluetooth Commands

Serial and Bluetooth use the same commands. Send inputs via a terminal (e.g., Arduino Serial Monitor or Bluetooth app) and receive system responses. Enter commands with a Newline or Carriage Return.

Command	Action	System Response	Notes
M	Show main menu	Settings Menu: 1. Alarm Time [06:00] 2. Run Duration [60 sec] 3. Retries [3] 4. Retry Interval [1 min] 5. Interval [2 days] 6. Status 7. Admin Menu M. Menu	Starts menu navigation
1	Set alarm time	Enter Alarm Time (HH:MM):	Enter HH:MM (e.g., 06:00)
HH:MM	Save alarm time	Alarm Set to HH:MM or Use HH:MM (Invalid)	Valid: 00:00–23:59; invalid shows error
2	Set run duration	11. Seconds 12. Minutes	Choose 11 or 12
11	Set seconds	Enter Seconds (1–60):	Enter 1–60 (e.g., 10) -(X)
X	Save seconds	Set X sec or Use 1–60(Invalid)	Sets run duration
12	Set minutes	Enter Minutes (1–5):	Enter 1–5 (e.g., 2)

X	Save minutes	Set X min or Use 1–5(Invalid)	Sets run duration (X * 60s)
3	Set retries	Enter Retries (0–5):	Enter 0–5 (e.g., 2) - (Y)
Y	Save retries	Retries: Y or Use 0–5(Invalid)	Sets Scheduled Mode retries
4	Set retry interval	Enter Retry Interval (1–60 min):	Enter 1–60 (e.g., 5) - (Z)
Z	Save retry interval	Retry Interval: Z min or Use 1–60 (Invalid)	Sets retry wait time
5	Set alarm interval	Enter Interval (1–2 days):	Enter 1–2 (e.g., 1) - (W)
W	Save alarm interval	Interval: W days or Use 1–2(Invalid)	Sets days between runs
6	Show status	Time: 12:00 Water: YES Tank: NO Motor: OFF Mode: AUTOMATIC	Shows sensor, motor, mode status
7	Access Admin Menu	Enter Password:	Enter Enarxi0000
Enarxi0000	Grant admin access	Admin Granted Admin Menu: 21. Set RTC 22. Manual Mode 23. Scheduled Mode 24. Reset M. Menu	Shows admin options

wrong password	Deny admin access	Wrong Password	Re-enter password
21	Set RTC time	Enter Date/Time (DD/MM/YYYY, HH:MM:SS):	Enter 08/06/2025, 12:30:00
DD/MM/YYYY, HH:MM:SS	Save RTC time	RTC Updated or Use DD/MM/YYYY, HH:MM:SS (Invalid) or No RTC	Sets system clock
22	Toggle Manual motor	Manual Motor ON/OFF	Toggles motor in Manual Mode
23	Toggle Scheduled Mode	Scheduled ON/OFF or No RTC	Toggles Scheduled Mode
24	Reset system	System Reset	Restores defaults, Auto Mode, Motor Off

Key System Logs (Serial/Bluetooth)

These messages appear automatically during operation (e.g., Scheduled Mode, button actions):

Log Message	Context
No RTC	RTC not detected
System Reset	System reset
Scheduled ON/OFF	Scheduled Mode toggled

Manual Motor ON/OFF	Manual motor toggled
Auto/Manual Mode	Mode switched
Sch Run STARTED	Scheduled run begins
Run Complete	Scheduled run ends
Starting Retries	Retries begin
Water Detected	Water present during retry
Retry Y Started	Retry Y begins
Retry Y OK/FAILED	Retry Y succeeds/fails
Retry Wait Started	Retry wait begins
Retries FAILED	All retries fail
Retry SUCCESS	Water Detected , Motor Starts
Motor STOPPED	Scheduled mode , when water stops or tank full

LCD Messages

The LCD shows system status, modes, and events. Messages are centered (16 characters) and alternate every 5 seconds in Scheduled Mode for some processes. Events display for 5 seconds before reverting.

Message	Context
ENARXI WATER REFILL SYS	System boot (5s)
Auto Mode	
HH:MM AUTO MOTOR ON/OFF	Current time, motor state
Manual Mode	
HH:MM MANUAL MOTOR ON/OFF	Current time, motor state
Scheduled Mode	
A HH:MM XS-Y-ZM-WD	Alarm time, settings (e.g., 60S-3-1M-2D) - (Default)
A HH:MM Sch Run STARTED	Run begins (alternates, 5s)
A HH:MM Run Complete	Run ends (alternates, 5s)
A HH:MM	Retries begin (alternates, 5s)

Starting Retries	
A HH:MM Water Detected	Water during retry (alternates, 5s)
A HH:MM Retry Y	Retry Y starts (alternates, 5s)
A HH:MM Retry Wait Started	Retry wait (alternates, 5s)
A HH:MM Retry FAILED	Retry fails (alternates, 5s)
A HH:MM Retries FAILED	All retries fail (alternates, 5s)
A HH:MM Retry SUCCESS	Retry succeeds (alternates, 5s)
A HH:MM Motor STOPPED	Scheduled mode stops (alternates, 5s)
Events/Errors	Displayed in Row 2
System Started	System initialized (5s)
RTC Not Found	RTC unavailable (5s)

System Reset	Reset triggered (5s)
Process Initiated	Button pressed (5s)
Scheduled ON/OFF	Scheduled Mode toggled (5s)
Manual Motor ON/OFF	Manual motor toggled (5s)
Auto Mode ON	Auto Mode entered (5s)
Manual Mode ON	Manual Mode entered (5s)
Sch Mode Disabled	No RTC for Scheduled Mode (5s)
Menu Displayed	Menu accessed (5s)
Set Alarm Time	Alarm input prompted (5s)
Alarm Set	Alarm saved (5s)
Invalid Alarm	Invalid alarm input (5s)
Set Duration	Duration type selected (5s)
Set Seconds	Seconds input prompted (5s)
Set X sec	Seconds saved (5s)
Invalid Seconds	Invalid seconds (5s)

Set Minutes	Minutes input prompted (5s)
Set X min	Minutes saved (5s)
Invalid Minutes	Invalid minutes (5s)
Set Retries	Retries input prompted (5s)
Set Y retries	Retries saved (5s)
Invalid Retries	Invalid retries (5s)
Set Retry Interval	Retry interval prompted (5s)
Set Z min retry	Retry interval saved (5s)
Invalid Retry Int	Invalid retry interval (5s)
Set Interval	Alarm interval prompted (5s)
Set W days	Alarm interval saved (5s)
Invalid Days	Invalid days (5s)
Status Displayed	Status shown (5s)
Enter Password	Password prompted (5s)
Admin Granted	Admin access granted (5s)

Wrong Password	Invalid password (5s)
Set RTC	RTC input prompted (5s)
RTC Updated	RTC saved (5s)
Invalid RTC	Invalid RTC input (5s)
RTC Not Available	RTC unavailable (5s)
Invalid Option	Invalid menu input (5s)